

Trigger Board Assembly

System documentation: pin-by-pin reference, block diagram, board interconnect and bill of materials

Prepared 2026-10-06 from the released KiCad projects (Base_Board rev "pin version", Logic_Board rev 0.2). No design changes are implied by this document.

1. System overview

The Trigger Board Assembly is two stacked 1.6 mm boards: the Base board has 6 copper layers (F.Cu, In1-In4, B.Cu) and the Logic board has 4 (F.Cu, In1, In2, B.Cu). The Logic board (150 x 39 mm) plugs onto the Base board through two 2x9 connectors: Base J8 (socket) mates with Logic J1 (pin header), and Base J2 (socket) mates with Logic J2 (pin header). Each board has its own 24 V to 5 V DC-DC converter.

Functional summary, derived from the schematic netlist (verify against design intent):

- Trigger path: an external trigger enters on BNC J6 or fiber header J10 (JP2 selects), optionally terminated in 50 ohm (JP3), and goes to the Logic board as Trig_In on J8 pin 1. The output pulse is driven by IC1 (IXDN614PI) as Trig_Out onto BNC J5 or fiber header J7 (JP1 selects; JP4 adds a 50 ohm load).
- Gating: three opto-isolated external inhibit inputs (J1 pins 7, 9, 11) and the Logic board decision signal In_from_lgboard are ANDed in U6 to form the IC1 input. JP5 and JP6 can force two of the inhibit channels to 5 V.
- Timing: U4 (74LS221) generates Master_TO_Out and Slave_TO_Out. Unit 1 uses C12 100 uF with R14 22 k (about 1.5 s, estimated $0.7 \times R \times C$); unit 2 uses C4 1 nF with R13 1.5 k (about 1 us). U9 (74AHCT123) stretches the output pulse into Trig_Sts using C13 10 uF and R31 100 k (about 0.45 s, estimated $0.45 \times R \times C$).
- Status: Remote, Local, Armed, OneShot, Normal, Power, Trigger and inhibit-summary states are passed through quad optocouplers (U2, U7, U8) to the isolated terminal block J3, powered from the external 24 V (EXT_24V).
- Mode selection: base switches SW2 and SW3 and push button SW1 drive the switch1/switch2 lines, the Arm/one-shot timer input and the Remote status net toward the Logic board.
- Logic board: NAND/AND/OR logic (U2-U5 SN74ACT00, U6 74AHCT04, U7 74LS08, U8 4075) forms the armed / one-shot / normal / remote-local state logic and returns In_from_lgboard and OR_Out.

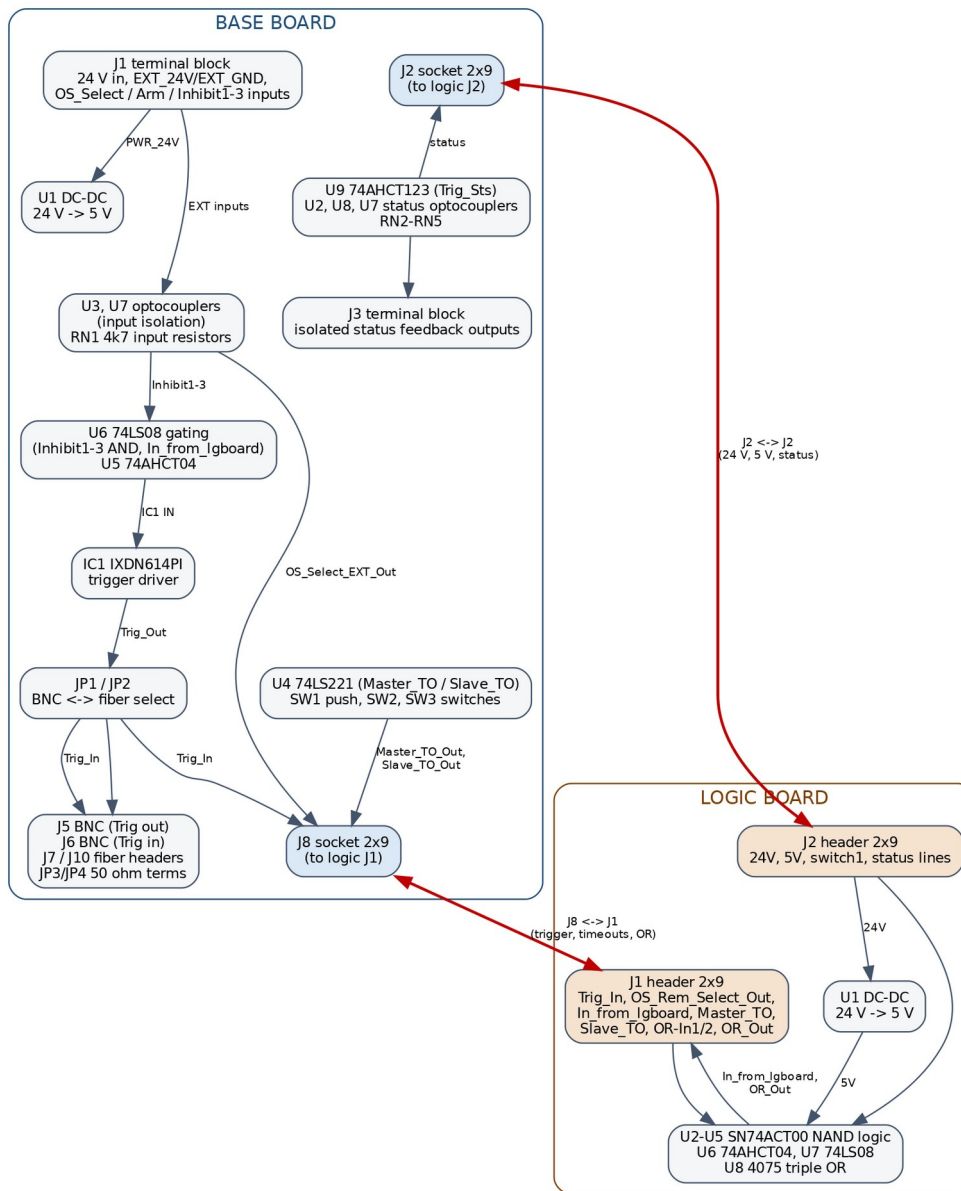


Figure 1. System block diagram (base + logic). Source: block_diagram.png in this folder.

2. Board-to-board interconnect

Base J8 plugs into Logic J1 and Base J2 plugs into Logic J2 (confirmed by the designer). Pins mate n to n; net names match on every populated pin of J8/J1, which supports this.

2.1 Base J8 to Logic J1

Pin	Base J8 net	Logic J1 net	Note
1	Trig_In	Trig_In	
2	GND	GND	
3	OS_Select_EXT_Out	OS_Rem_Select_Out	name differs
4	In_from_Igboard	In_from_Igboard	
5	Master_TO_Out	Master_TO	
6	GND	GND	

7	GND	GND	
8	Slave_TO_Out	Slave_TO	
9	NC	NC	spare
10	NC	NC	spare
11	NC	NC	spare
12	NC	NC	spare
13	NC	NC	spare
14	NC	NC	spare
15	NC	NC	spare
16	OR-In1	OR-In1	
17	OR-In2	OR-In2	
18	OR_Out	OR_Out	

2.2 Base J2 to Logic J2

Pin	Base J2 net	Logic J2 net	Note
1	PWR_24V	24V	
2	GND	GND	
3	GND	GND	
4	NC	NC	spare
5	lg_5V_sts	5V	name differs
6	switch2pin	Remote Sts	name differs
7	switch1pin	switch1pin	
8	switch1opt	switch1opt	
9	NC	NC	spare
10	NC	NC	spare
11	NC	NC	spare
12	NC	NC	spare
13	NC	NC	spare
14	Normal_Sts	Normal_Sts	
15	OneShot_Sts	OneShot_Sts	
16	Armed_Sts	Armed_Sts	
17	Local_Sts	Local_Sts	
18	Remote Sts	Remote Sts	

Notes on differing names (J2): pin 1 is the 24 V rail named PWR_24V on the base and 24V on the logic board. Pin 5 is the logic board 5 V rail, read on the base as lg_5V_sts (5 V present sense). Pin 6 is switch2pin on the base (SW2 output) and lands on the Logic Remote Sts net, which is also on pin 18; so SW2 on the base (the remote/local selector) drives the Logic board Remote Sts net, as intended (confirmed by the designer). Naming differences on J8 (Master_TO_Out vs Master_TO, Slave_TO_Out vs Slave_TO, OS_Select_EXT_Out vs OS_Rem_Select_Out, GND vs /GND) are the same signals.

3. Base board pinouts

3.1 J1: Field terminal block (Phoenix Contact 1700812, 18 positions): 24 V power, external isolated 24 V and inputs

Pin	Function / pad name	Net
1	1	PWR_24V
2	2	GND
3	3	EXT_24V
4	4	EXT_GND

5	5	OS_Select_EXT
6	6	Arm_OS_EXT
7	7	Inhibit1_EXT
8	8	NC
9	9	Inhibit2_EXT
10	10	Inhibit2_EXT_GND
11	11	Inhibit3_EXT
12	12	Inhibit3_EXT_GND
13	13	NC
14	14	NC
15	15	NC
16	16	NC
17	17	NC
18	18	NC

3.2 J3: Field terminal block (Phoenix Contact 1700812, 18 positions): isolated status outputs

Pin	Function / pad name	Net
1	1	NC
2	2	NC
3	3	Rmt_Sts_FB
4	4	Local_Sts_FB
5	5	Inhib_Summ_FB
6	6	Armed_Sts_FB
7	7	Normal_Sts_FB
8	8	OneShot_Sts_FB
9	9	NC
10	10	Pwr_Sts_FB
11	11	NC
12	12	Trg_Sts_FB
13	13	NC
14	14	Inhibit1_Sts_FB
15	15	Inhibit2_Sts
16	16	Inhibit2_Sts_FB
17	17	Inhibit3_Sts
18	18	Inhibit3_Sts_FB

3.3 J2: 2x9 socket to Logic J2

Pin	Function / pad name	Net
1	Pin_1	PWR_24V
2	Pin_2	GND
3	Pin_3	GND
4	Pin_4	NC
5	Pin_5	Ig_5V_sts
6	Pin_6	switch2pin
7	Pin_7	switch1pin
8	Pin_8	switch1opt
9	Pin_9	NC
10	Pin_10	NC
11	Pin_11	NC
12	Pin_12	NC
13	Pin_13	NC
14	Pin_14	Normal_Sts
15	Pin_15	OneShot_Sts
16	Pin_16	Armed_Sts
17	Pin_17	Local_Sts

18	Pin_18	Remote Sts
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3.4 J8: 2x9 socket to Logic J1

Pin	Function / pad name	Net
1	Pin_1	Trig_In
2	Pin_2	GND
3	Pin_3	OS_Select_EXT_Out
4	Pin_4	In_from_Igboard
5	Pin_5	Master_TO_Out
6	Pin_6	GND
7	Pin_7	GND
8	Pin_8	Slave_TO_Out
9	Pin_9	NC
10	Pin_10	NC
11	Pin_11	NC
12	Pin_12	NC
13	Pin_13	NC
14	Pin_14	NC
15	Pin_15	NC
16	Pin_16	OR-In1
17	Pin_17	OR-In2
18	Pin_18	OR_Out

3.5 J5: BNC, trigger output

Pin	Function / pad name	Net
1	In	(J5-In)
2	Ext	GND

3.6 J6: BNC, trigger input

Pin	Function / pad name	Net
1	In	(J6-In)
2	Ext	GND

3.7 J7: 3-pin header, fiber transmitter (output side) connection

Pin	Function / pad name	Net
1	Pin_1	(J7-Pin_1)
2	Pin_2	GND
3	Pin_3	5V

3.8 J10: 3-pin header, fiber receiver (input side) connection

Pin	Function / pad name	Net
1	Pin_1	(J10-Pin_1)
2	Pin_2	GND
3	Pin_3	5V

J7/J10 fiber header assignments are inferred from jumper wiring (JP1 B to J7 pin 1; JP2 B to J10 pin 1): J7 carries the output trigger, J10 the input trigger. Pin 1 = signal, pin 2 = GND, pin 3 = 5 V.

3.9 Jumpers and switches

Ref	Pins and nets
JP1	1(A)=(J5-In); 2(C)=Trig_Out; 3(B)=(J7-Pin_1)
JP2	1(A)=(J6-In); 2(C)=Trig_In; 3(B)=(J10-Pin_1)
JP3	1(A)=(J6-In); 2(B)=(JP3-B)
JP4	1(A)=(JP4-A); 2(B)=(J5-In)
JP5	1(A)=5V; 2(B)=(JP5-B)
JP6	1(A)=5V; 2(B)=(JP6-B)
SW1	1(1)=(RN5A-R1.1); 2(2)=(U7-CT1)
SW2	1(B)=switch2pin; 2(C)=GND; 3(A)=5V
SW3	1(B)=switch1pin; 2(C)=GND; 3(A)=switch1opt

4. Logic board pinouts

4.1 J1: 2x9 pin header to Base J8

Pin	Net
1	Trig_In
2	GND
3	OS_Rem_Select_Out
4	In_from_Igboard
5	Master_TO
6	GND
7	GND
8	Slave_TO
9	NC
10	NC
11	NC
12	NC
13	NC
14	NC
15	NC
16	OR-In1
17	OR-In2
18	OR_Out

4.2 J2: 2x9 pin header to Base J2

Pin	Net
1	24V
2	GND
3	GND
4	NC
5	5V
6	Remote_Sts
7	switch1pin
8	switch1opt
9	NC
10	NC
11	NC
12	NC
13	NC
14	Normal_Sts
15	OneShot_Sts

16	Armed_Sts
17	Local_Sts
18	Remote Sts

5. Integrated circuits: pin-by-pin

Net names in parentheses are local nets named after a pin they connect to. NC = no connection.

Base IC1: IXDN614PI trigger driver (OUT1/OUT2 paralleled)

Pin	Function	Net
1	VCC_1	5V
2	IN	(IC1-IN)
3	NC	NC
4	GND_1	GND
5	GND_2	GND
6	OUT_1	Trig_Out
7	OUT_2	Trig_Out
8	VCC_2	5V

Base U1: WPME-FDSM DC-DC, 24 V to 5 V

Pin	Function	Net
1	1	PWR_24V
2	2	GND
3	3	5V

Base U2: PS2501L-4 optocoupler: status outputs

Pin	Function	Net
1	A1	(RN3D-R4.1)
2	C1	GND
3	A2	(RN3E-R5.1)
4	C2	GND
5	A3	(RN4A-R1.1)
6	C3	GND
7	A4	(RN4B-R2.1)
8	C4	GND
9	ET4	OneShot_Sts_FB
10	CT4	(RN5B-R2.2)
11	ET3	Armed_Sts_FB
12	CT3	(RN2F-R6.2)
13	ET2	Local_Sts_FB
14	CT2	(RN2E-R5.2)
15	ET1	Rmt_Sts_FB
16	CT1	(RN2D-R4.2)

Base U3: PS2501L-4 optocoupler: external inputs (OS select, inhibits)

Pin	Function	Net
1	A1	(RN1A-R1.2)
2	C1	EXT_GND
3	A2	(RN1B-R2.2)
4	C2	EXT_GND
5	A3	(RN1C-R3.2)
6	C3	Inhibit2_EXT_GND
7	A4	(RN1D-R4.2)
8	C4	Inhibit3_EXT_GND

9	ET4	(JP5-B)
10	CT4	5V
11	ET3	(JP6-B)
12	CT3	5V
13	ET2	(RN6D-R4.2)
14	CT2	5V
15	ET1	GND
16	CT1	OS_Select_EXT_Out

Base U4: 74LS221 dual monostable

Pin	Function	Net
1	~{A}	GND
2	B	(RN5A-R1.1)
3	~{CLR}	5V
4	~{Q}	NC
5	Q	NC
6	Cext	(U4B-Cext)
7	RCext	(U4B-RCext)
8	GND	GND
9	~{A}	GND
10	B	Master_TO_Out
11	~{CLR}	5V
12	~{Q}	Slave_TO_Out
13	Q	Master_TO_Out
14	Cext	(U4A-Cext)
15	RCext	(U4A-RCext)
16	VCC	5V

Base U5: 74AHCT04 hex inverter

Pin	Function	Net
1		(RN6D-R4.2)
2		(RN3A-R1.2)
3		(JP6-B)
4		(RN3B-R2.2)
5		(JP5-B)
6		(RN3C-R3.2)
7	GND	GND
8		OR-In2
9		Ig_5V_sts
10		OR-In1
11		5V
12		Pwr_Sts
13		OR_Out
14	VCC	5V

Base U6: 74LS08 quad AND

Pin	Function	Net
1		(RN6D-R4.2)
2		(JP6-B)
3		(RN8B-R2.2)
4		(RN8B-R2.2)
5		(JP5-B)
6		(RN8C-R3.2)

7	GND	GND
8		Inhib_Sum
9		(RN8C-R3.2)
10		5V
11		(IC1-IN)
12		In_from_Igboard
13		(RN8C-R3.2)
14	VCC	5V

Base U7: PS2501L-4 optocoupler: Arm input and inhibit status

Pin	Function	Net
1	A1	(RN1E-R5.2)
2	C1	EXT_GND
3	A2	(RN3A-R1.1)
4	C2	GND
5	A3	(RN3B-R2.1)
6	C3	GND
7	A4	(RN3C-R3.1)
8	C4	GND
9	ET4	Inhibit3_Sts_FB
10	CT4	(RN2C-R3.2)
11	ET3	Inhibit2_Sts_FB
12	CT3	(RN2B-R2.2)
13	ET2	Inhibit1_Sts_FB
14	CT2	(RN2A-R1.2)
15	ET1	(RN5A-R1.1)
16	CT1	(U7-CT1)

Base U8: PS2501L-4 optocoupler: status outputs

Pin	Function	Net
1	A1	(RN4C-R3.1)
2	C1	GND
3	A2	(RN4D-R4.1)
4	C2	GND
5	A3	(RN4E-R5.1)
6	C3	GND
7	A4	(RN4F-R6.1)
8	C4	GND
9	ET4	Inhib_Summ_FB
10	CT4	(RN5F-R6.2)
11	ET3	Trg_Sts_FB
12	CT3	(RN5E-R5.2)
13	ET2	Pwr_Sts_FB
14	CT2	(RN5D-R4.2)
15	ET1	Normal_Sts_FB
16	CT1	(RN5C-R3.2)

Base U9: 74AHCT123 dual monostable

Pin	Function	Net
1	A	GND
2	B	Trig_Out
3	Clr	5V
4	~{Q}	NC
5	Q	NC
6	Cext	NC
7	RCext	NC

8	GND	GND
9	A	NC
10	B	NC
11	Clr	NC
12	~{Q}	NC
13	Q	Trig_Sts
14	Cext	GND
15	RCext	(U9A-RCext)
16	VCC	5V

Logic U1: WPME-FDSM DC-DC, 24 V to 5 V

Pin	Function	Net
1	1	24V
2	2	GND
3	3	5V

Logic U2: SN74ACT00

Pin	Function	Net
1		(U2-Pad1)
2		Master_TO
3		(U2-Pad3)
4		(U2-Pad3)
5		5V
6		(U2-Pad13)
7	GND	GND
8		(U2-Pad12)
9		Slave_TO
10		(U2-Pad10)
11		(U2-Pad10)
12		(U2-Pad12)
13		(U2-Pad13)
14	VCC	5V

Logic U3: SN74ACT00

Pin	Function	Net
1		(U2-Pad12)
2		Trig_In
3		(U3-Pad3)
4		(U3-Pad4)
5		Trig_In
6		(U3-Pad12)
7	GND	GND
8		(U3-Pad4)
9		(U2-Pad12)
10		5V
11		(U3-Pad11)
12		(U3-Pad12)
13		Master_TO
14	VCC	5V

Logic U4: SN74ACT00

Pin	Function	Net
1		(U3-Pad3)
2		(U4-Pad2)
3		(U4-Pad3)
4		(U4-Pad3)
5		(U4-Pad5)
6		(U4-Pad2)
7	GND	GND
8		(U4-Pad5)
9		(U3-Pad11)
10		5V
11		(U4-Pad11)
12		Trig_In
13		5V
14	VCC	5V

Logic U5: SN74ACT00

Pin	Function	Net
1		(U4-Pad3)
2		(U4-Pad11)
3		(U5-Pad3)
4		(U4-Pad2)
5		(U4-Pad11)
6		(U5-Pad13)
7	GND	GND
8		Armed_Sts
9		(U5-Pad3)
10		(U2-Pad1)
11		(U2-Pad1)
12		Armed_Sts
13		(U5-Pad13)
14	VCC	5V

Logic U6: 74AHCT04

Pin	Function	Net
1		Remote Sts
2		Local_Sts
3		OneShot_Sts
4		Normal_Sts
5		OS_Rem_Select_Out
6		(U6-Pad6)
7	GND	GND
8		(U6-Pad8)
9		switch1pin
10		(U6-Pad10)
11		Remote Sts
12		switch1opt
13		Remote Sts
14	VCC	5V

Logic U7: 74LS08

Pin	Function	Net
1		Remote Sts
2		(U6-Pad6)
3		(U7-Pad3)
4		(U6-Pad6)
5		(U6-Pad8)
6		(U7-Pad6)
7	GND	GND
8		(U7-Pad8)
9		(U6-Pad8)
10		(U6-Pad10)
11		In_from_Igboard
12		Trig_In
13		(U7-Pad13)
14	VCC	5V

Logic U8: 4075 triple OR

Pin	Function	Net
1		(U7-Pad3)
2		(U7-Pad6)
3		switch1pin
4		Armed_Sts
5		OS_Rem_Select_Out
6		(U7-Pad13)
7	VSS	GND
8		(U7-Pad8)
9		OneShot_Sts
10		OR_Out
11		OR-In1
12		OR-In2
13		GND
14	VDD	5V

For 74xx devices, the gate-level pin functions follow the standard pinouts; the table lists the nets as routed.

6. Passive networks

Board	Ref	Value	Pin nets
Base	C1	C_Polarized_US	1=PWR_24V; 2=GND
Base	C10	100n	1=5V; 2=GND
Base	C11	100n	1=GND; 2=5V
Base	C12	100u	1=(U4A-Cext); 2=(U4A-RCext)
Base	C13	10u	1=(U9A-RCext); 2=GND
Base	C2	C_Polarized_US	1=5V; 2=GND
Base	C3	100n	1=GND; 2=5V
Base	C4	1n	1=(U4B-Cext); 2=(U4B-RCext)
Base	C5	100n	1=(RN5A-R1.1); 2=GND
Base	C6	100n	1=GND; 2=5V
Base	C7	100n	1=GND; 2=5V

Base	R13	1k5	1=(U4B-RCext); 2=5V
Base	R14	22k	1=(U4A-RCext); 2=5V
Base	R16	100	1=(U7-CT1); 2=5V
Base	R30	50_1W	1=(JP4-A); 2=GND
Base	R31	100k	1=(U9A-RCext); 2=5V
Base	R32	50_1W	1=(JP3-B); 2=GND
Base	RN1	4k7	1=OS_Select_EXT; 2=(RN1A-R1.2); 3=Inhibit1_EXT; 4=(RN1B-R2.2); 5=Inhibit2_EXT; 6=(RN1C-R3.2); 7=Inhibit3_EXT; 8=(RN1D-R4.2); 9=Arm_OS_EXT; 10=(RN1E-R5.2)
Base	RN2	1k	1=EXT_24V; 2=(RN2A-R1.2); 3=Inhibit2_Sts; 4=(RN2B-R2.2); 5=Inhibit3_Sts; 6=(RN2C-R3.2); 7=EXT_24V; 8=(RN2D-R4.2); 9=EXT_24V; 10=(RN2E-R5.2); 11=EXT_24V; 12=(RN2F-R6.2)
Base	RN3	470	1=(RN3A-R1.1); 2=(RN3A-R1.2); 3=(RN3B-R2.1); 4=(RN3B-R2.2); 5=(RN3C-R3.1); 6=(RN3C-R3.2); 7=(RN3D-R4.1); 8=Remote Sts; 9=(RN3E-R5.1); 10=Local_Sts
Base	RN4	470	1=(RN4A-R1.1); 2=Armed_Sts; 3=(RN4B-R2.1); 4=OneShot_Sts; 5=(RN4C-R3.1); 6=Normal_Sts; 7=(RN4D-R4.1); 8=Pwr_Sts; 9=(RN4E-R5.1); 10=Trig_Sts; 11=(RN4F-R6.1); 12=Inhib_Sum
Base	RN5	1k	1=(RN5A-R1.1); 2=GND; 3=EXT_24V; 4=(RN5B-R2.2); 5=EXT_24V; 6=(RN5C-R3.2); 7=EXT_24V; 8=(RN5D-R4.2); 9=EXT_24V; 10=(RN5E-R5.2); 11=EXT_24V; 12=(RN5F-R6.2)
Base	RN6	10k	1=GND; 2=NC; 3=OR_Out; 4=(JP6-B); 5=(RN6D-R4.2)
Base	RN7	10k	1=GND; 2=(RN6D-R4.2); 3=(JP6-B); 4=(JP5-B); 5=Ig_5V_sts
Base	RN8	10k	1=GND; 2=(JP5-B); 3=(RN8B-R2.2); 4=(RN8C-R3.2); 5=In_from_Igboard
Logic	C1	C_Polarized_US	1=5V; 2=GND
Logic	C2	C_Polarized_US	1=24V; 2=GND
Logic	C3	100n	1=GND; 2=5V
Logic	C4	100n	1=GND; 2=5V
Logic	C5	100n	1=GND; 2=5V
Logic	C6	100n	1=GND; 2=5V
Logic	C7	100n	1=GND; 2=5V
Logic	C8	100n	1=GND; 2=5V
Logic	C9	100n	1=GND; 2=5V
Logic	R1	10k	1=Remote Sts; 2=OS_Rem_Select_Out

7. Bill of materials

Full BOM, with base and logic breakouts, is in Trigger_Board_System_BOM.xlsx. System totals below (R: ref prefix B = Base, L = Logic).

Qty	Ref(s)	Schematic value	Description	MFR part #	Mouser #
2	B:C1, B:C2	C_Polarized_US	100uF 50V Electrolythic Capacitor	ESL107M050AGMAA	80-ESL107M050AGMAA
6	B:C3, B:C5, B:C6, B:C7, B:C10, B:C11	100n	100nF Ceramic disk capacitor	S104Z93Z5VL83LOR	594-S104Z93Z5VL83LOR
1	B:C4	1n	1nF Ceramic Disk Capacitor	FA28C0G1H102JNU00	810-FA28C0G1H10200
1	B:C12	100u	100uF 25V Electrolythic Capacitor	Kit part (no part number)	NA

1	B:C13	10u	10uF Ceramic Disk Capacitor	FG11X7R1H106KRT00	810-FG11X7R1H106KRT0
1	B:IC1	IXDN614PI	IXYS MOSFET Gate Driver	IXDN614PI	849-IXDN614PI
1	B:J1	J6	Screw Terminal Blocks	1725041	651-1725041
2	B:J2, B:J8	Conn_02x09_Counter_Clockwise	02x09 Pin Socket	215307-9	571-215307-9
1	B:J3	1700812	Screw Terminal Blocks	1725041	651-1725041
2	B:J5, B:J6	Conn_Coaxial	BNC female PCB mounted Connector	1-1478204-0	571-1-1478204-0
2	B:J7, B:J10	Conn_01x03_Pin	Fiber RX/TX Eval boards.	AFBR-0548Z	630-AFBR-0548Z
2	B:JP1, B:JP2	Jumper_3_Open	Jumper Pins (2 pin ones must be modded) - 6 used + 1 spare	90120-0763	538-90120-0763
4	B:JP3, B:JP4, B:JP5, B:JP6	Jumper_2_Small_Open	Jumper Pins (2 pin ones must be modded) - 6 used + 1 spare	90120-0763	538-90120-0763
1	B:R13	1k5	1K5 ohm 1/4W	MFR-25FBF52-1K5	603-MFR-25FBF52-1K5
1	B:R14	22k	22Kohm 1/4W	MFR-25FTF52-22K	603-MFR-25FTF52-22K
1	B:R16	100	100 ohm 1/4W	MFR-25FBF52-100R	603-MFR-25FBF52-100R
2	B:R30, B:R32	50_1W	50-Ohm 1W Resistor	PAC100005009FA1000	71-PAC100005009FA100
1	B:R31	100k	100K ohm 1/4W	MFR-25FTE52-100K	603-MFR-25FTE52-100K
1	B:RN1	4k7	Resistor Network Isolated(4k7x05)	770103472P	774-770103472P
2	B:RN2, B:RN5	1k	Resistor Network Isolated(1k06)	4612X-102-102LF	652-4612X-2LF-1K
1	B:RN3	470	Resistor Network Isolated (470x05)	4610X-102-471LF	652-4610X-2LF-470
1	B:RN4	470	Resistor Network Isolated (470x06)	4612X-102-471LF	652-4612X-2LF-470
3	B:RN6, B:RN7, B:RN8	10k	Resistor Network Bussed(10Kx04)	4605X-101-103FLF	652-4605X-101-103FLF
1	B:SW1	SW_Push	Tactile switch	TS02-66-60-BK-260-LCR-D	179-TS026660BK260LCR
2	B:SW2, B:SW3	SW_Wuerth_450301014042	Wuerth Switch	450301014042	710-450301014042
2	B:U1, L:U1	WPME-FDSM_SIP-3	Wuerth Elektronik Voltage Regulator	173010578	710-173010578
4	B:U2, B:U3, B:U7, B:U8	PS2501L-4-A	4-Channel OptoCoupler	PS2501-4-A	551-PS2501-4-A
1	B:U4	74LS221	Texas Instruments Monostable Multivibrator	SN74LS221NE4	595-SN74LS221NE4
2	B:U5, L:U6	74AHCT04	Texas Instruments Hex Inverter	SN74AHCT04N	595-SN74AHCT04N
1	B:U6	74LS08	Texas Instruments 4-Gate AND	SN74LS08N	595-SN74LS08N
1	B:U9	74AHCT123	Texas Instruments Monostable Multivibrator	SN74AHCT123AN	595-SN74AHCT123AN
2	L:C1, L:C2	C_Polarized_US	100uF 50V Electrolytic Capacitor	ESL107M050AGMAA	80-ESL107M050AGMAA
7	L:C3, L:C4, L:C5, L:C6, L:C7, L:C8, L:C9	100n	100 nF Ceramic disk capacitor	S104Z93Z5VL83LOR	594-S104Z93Z5VL83LOR
2	L:J1, L:J2	Conn_02x09_Counter_Clockwise	02x09 Pin Header	826925-9	571-826925-9
1	L:R1	10k	10K 1% 1/4W Resistor	Kit part (no part number)	NA
4	L:U2, L:U3, L:U4, L:U5	SN74ACT00	Texas Instruments Quad 2-Input NAND	SN74ACT00N	595-SN74ACT00N
1	L:U7	74LS08	Texas Instruments 4-Gate AND	SN74LS08N	595-SN74LS08N
1	L:U8	4075	Texas Instruments Triple 3-input OR Gate	CD4075BE	595-CD4075BE

Not on the PCBs (see the Off-Board Items sheet of the BOM spreadsheet): the AFBR-0548Z fiber RX/TX evaluation board (qty 1, plugs into J7 and J10, modified), Phoenix Contact housing parts (2278500, 2278526, 2278542, 2 x 2896238), 6 jumper shunts (JM-2BK-61) and the bare PCBs. The source of part numbers is Final_Trigger_Board_BOM.xlsx.

Total: 72 parts (52 base, 20 logic), excluding bare PCBs, shunts, sockets and hardware.

8. BOM reconciliation and open items

Part numbers above come from Final_Trigger_Board_BOM.xlsx matched by reference designator. Every schematic reference appears in that BOM. Differences found:

- C4 and C13: the original Final BOM had the designators swapped. The board is correct (C4 marked 102 = 1 nF, C13 marked 106 = 10 uF, as in the schematic) and Final_Trigger_Board_BOM.xlsx is now corrected.
- C12: the fitted part is a regular 100 uF 25 V electrolytic from a capacitor kit (no part number), measured D 5.18 mm, H 11.37 mm, lead pitch 1.77 mm as bent, with its leads bent so the top PCB does not hit it. This matches the schematic. The original Final BOM MPN reads as 2.2 uF / 100 V and was replaced with kit part.
- Logic R1: the fitted resistor is 10 k, 1% (bands brown-black-black-red-brown), from a resistor kit with no part number. This matches the schematic. The original Final BOM MPN reads as 6.8 k and was replaced with kit part.
- The original Final BOM listed JP7 and JP10. These are the 3-pin 2.54 mm positions where the eval board pins are soldered, i.e. J7 and J10 in the schematic, not extra jumpers; they are not jumpers. There are 6 jumpers (JP1-JP6); the Final BOM quantity of 7 is 6 used plus 1 spare, so item 10 reads JP1-JP6, quantity 7 (6 + 1 spare).
- Logic U1 Mouser number was corrected to 710-173010578 to match base U1, and base U5 to 595-SN74AHCT04N (missing prefix).
- One AFBR-0548Z evaluation board (qty 1) is listed for both J7 and J10. The bent interface pins were removed and replaced with straight pins facing vertically down (longer part of the pins on the side opposite the fiber ports); the replacement pins are generic 3-position, 2.54 mm pitch header (no part number) cut from a longer strip on hand.
- Terminal blocks J1 and J3: the BOM lists 6 blocks of MPN 1725041 for 18 positions each, which resolves the earlier library-text/footprint question if that part is 3-position (not verified).
- Schematic symbol U2/U3/U7/U8 is PS2501L-4-A; the ordered part is PS2501-4-A in a DIP-16 footprint.
- Mating of J8/J1 and J2/J2 is confirmed by the designer (section 2). Functional descriptions in section 1 are derived from the netlist, not from design intent.

8.1 Assembly notes

- Fiber evaluation board (AFBR-0548Z): the bent interface pins were removed and replaced with straight pins facing vertically down, the longer part of the pins on the side opposite the fiber ports.
- Logic R1 (10 k, 1%): a resistor from a resistor kit (bands brown-black-black-red-brown), no part number.
- C12 (100 uF, 25 V): a regular electrolytic capacitor from an Amazon capacitor kit (no part number; D 5.18 mm, H 11.37 mm, lead pitch 1.77 mm as bent), with its leads bent so the top (logic) board does not hit it.

8.2 Final testing (from the Final Testing sheet)

Two assemblies were tested against 22 items. All items passed on both assemblies, including BNC and fiber input/output, inhibit 1-3, the one-shot button, local/remote and normal/one-shot switches, and the 50-ohm termination jumpers.

Test	Description	Assy 1	Assy 2
50-ohm termination jumpers	50 ohms across BNC connectors when jumper present. BNC open if jumper removed.	Pass	Pass
Local-Remote switch	Local and Remote status outputs toggle properly with switch input.	Pass	Pass
Normal-OS switch	Normal and OS feedbacks are triggered appropriately.	Pass	Pass
Power status feedback	HIGH(24V) when PWR and EXT PWR are present	Pass	Pass
Inhibit Summation Status (inh2 jumper)	Triggered on inh2 jumper	Pass	Pass
Inhibit Summation Status (inh3 jumper)	Triggered on inh3 jumper	Pass	Pass

Inhibit 1 feedback	triggers HIGH if inhibit 1 input LO and viceversa	Pass	Pass
input selection jumper	Propely labeled and selects the appropriate input	Pass	Pass
output selection jumper	Propely labeled and selects the appropriate output	Pass	Pass
Inhibit 2 jumper	Stops Pulses and reflects properly on INH summation	Pass	Pass
Inhibit 3 jumper	Stops Pulses and reflects properly on INH summation	Pass	Pass
Trigger_sts_fb	tested on BNC triggering	Pass	Pass
OS button	allows one pulse when proper switch setup and button pressede	Pass	Pass
inhibit 2 in/out	inhibit 2 input signal propely closes the selected voltage on INH2 feedback.	Pass	Pass
inhibit 3 in/out	inhibit 3 input signal propely closes the selected voltage on INH3 feedback.	Pass	Pass
Remote OS signal input	When swtiches are appropriate, remote OS is selected and Remote OS Signal is received a single pulse is observed at the output	Pass	Pass
BNC input	Input works properly	Pass	Pass
BNC output	Output works properly	Pass	Pass
Fiber input	Input works properly	Pass	Pass
fiber output	Output works properly	Pass	Pass
INH1 stops pulses	When pulsing, INH1 LOW stops the pulses.	Pass	Pass
INH2 stops pulses	When Pulsing INH2 LOW stops the pulses.	Pass	Pass
INH3 stops pulses	When Pulsing INH3 LOW stops the pulses.	Pass	Pass

9. Gerber verification (released Gerbers vs KiCad project files)

Checked 2026-10-06 by parsing the released Gerber/drill zips (generated 2025-05-30 by KiCad 8.0.8) and comparing them with Base_Board/KiCad/baseboard.kicad_pcb and Logic_Board/KiCad/logicboard.kicad_pcb. KiCad itself was not available, so the check compares geometry; it is not a byte-for-byte regeneration.

Item	Base board	Logic board
Copper tracks, every layer	all 966 match exactly (position, width)	all 369 match exactly
Zone fills	22 of 22 identical	3 of 3 present (count match)
Drill holes (PTH)	389 of 389 within 1 um; no NPTH holes in either file	173 of 173 exact
Board outline	148 lines + 48 arcs, count match	4 lines, count match
Solder-mask openings	352 flashes, all at pad positions	157 flashes, all at pad positions
Not verified	silkscreen, paste, pad/aperture sizes and shapes	same

Result: copper, drills, outline and mask openings in the released Gerbers match the current project files. The base PCB file was last saved 2025-06-17, after the 2025-05-30 Gerbers, but no geometry difference was found, so those later saves did not change the checked items.